UP Board Class 12 Biology - 348 (CI) 2023 Question Paper with Solutions

Time Allowed :3 Hours	**Maximum Marks :**100	**Total Questions :**9

General Instructions

Read the following instructions very carefully and strictly follow them:

1. There are in all nine questions in this question paper.
2. All questions are compulsory.
3. In the beginning of each question, the number of parts to be attempted are clearly mentioned.
4. Marks allotted to the questions are indicated against them.
5. Start solving from the first question and proceed to solve till the last one. Do not waste your time over a question you cannot solve.

Multiple Choice Type Questions

1.(a) Genetically modified plant is/are

(A) BT maize
(B) BT cotton
(C) BT rice
(D) All of these

Correct Answer: (D) All of these

Solution:
Step 1: Understanding genetically modified (GM) plants.
Genetically modified plants have specific genes inserted into their genome to enhance traits such as pest resistance, disease resistance, or nutritional quality. A common modification is the incorporation of the *Bacillus thuringiensis* (Bt) gene, which produces a protein toxic to certain insect pests but safe for humans.

Step 2: Analysis of options.
- (A) Bt maize: Genetically modified maize containing the Bt gene for insect resistance.
- (B) Bt cotton: One of the most widely cultivated GM crops, Bt cotton resists bollworm pests.
- (C) Bt rice: Developed through genetic engineering to resist stem borers.
- (D) All of these: Since Bt maize, Bt cotton, and Bt rice are all genetically modified, this is

the correct choice.

Step 3: Conclusion.
The correct answer is (D) All of these, as all listed plants are genetically modified varieties.

Quick Tip

Bt crops contain genes from *Bacillus thuringiensis* that make them resistant to insect pests, reducing the need for chemical pesticides.

(b) Which one of the following is a false fruit?

(A) Grape
(B) Mango
(C) Guava
(D) Apple

Correct Answer: (D) Apple

Solution:
Step 1: Understanding false fruit.
A false fruit (or pseudocarp) develops not only from the ovary but also from other floral parts such as the thalamus or receptacle. In contrast, a true fruit develops solely from the ovary of the flower.

Step 2: Analysis of options.
- (A) Grape: A true fruit, developing only from the ovary.
- (B) Mango: Also a true fruit, formed exclusively from the ovary.
- (C) Guava: A true fruit as well, arising solely from the ovary.
- (D) Apple: A false fruit, because the fleshy edible part develops mainly from the thalamus, not just the ovary.

Step 3: Conclusion.
Hence, the correct answer is (D) Apple, which is classified as a false fruit.

Quick Tip

False fruits develop from the ovary plus other floral parts, while true fruits originate only from the ovary.

(c) The number (X) of chromosomes in the endosperm of the angiospermic plant is

(A) 1X
(B) 2X
(C) 3X
(D) 4X

Correct Answer: (C) 3X

Solution:
Step 1: Understanding endosperm formation.
In angiosperms, double fertilization takes place, where one male gamete fuses with the egg cell to form the diploid zygote (2X), and the other male gamete fuses with the two polar nuclei present in the central cell of the embryo sac.

Step 2: Chromosome contribution.
- Each polar nucleus is haploid (1X), so together they contribute 2X.
- The male gamete is haploid (1X).
Thus, the endosperm formed is triploid (3X).

Step 3: Conclusion.
Therefore, the correct answer is (C) 3X, as the endosperm is typically triploid in angiosperms.

Quick Tip

In angiosperms, the endosperm is triploid (3X), formed by double fertilization, which is a unique feature of flowering plants.

(d) The name of pathogen of typhoid is

(A) Ascaris
(B) Amoeba
(C) Paramecium
(D) Salmonella

Correct Answer: (D) Salmonella

Solution:
Step 1: Understanding typhoid disease.
Typhoid fever is a bacterial infection transmitted primarily through contaminated food and

water. It mainly affects the intestinal tract and bloodstream.

Step 2: Identifying the pathogen.
- (A) Ascaris: A parasitic roundworm causing ascariasis, not typhoid.
- (B) Amoeba: Causes amoebic dysentery, not typhoid.
- (C) Paramecium: A free-living ciliate protozoan, non-pathogenic to humans.
- (D) Salmonella: Specifically, *Salmonella typhi* is the bacterium responsible for typhoid fever.

Step 3: Conclusion.
Hence, the correct answer is (D) Salmonella, as *Salmonella typhi* causes typhoid fever.

Quick Tip

Typhoid fever is caused by *Salmonella typhi* and is transmitted through contaminated food and water.

Subjective Questions

2.(a) Write in brief 'In Vitro Fertilization' (IVF).

Solution:
In Vitro Fertilization (IVF) is an assisted reproductive technology (ART) used to help couples conceive a child when natural conception is difficult. The term "in vitro" means "outside the body," and in this process, fertilization occurs in a laboratory dish rather than inside the woman's body.
Steps involved in IVF:

1. **Ovarian Stimulation:** Hormonal injections are given to stimulate the ovaries to produce multiple eggs.
2. **Egg Retrieval:** Mature eggs are collected from the woman's ovaries through a minor surgical procedure.
3. **Fertilization:** The eggs are combined with sperm in a controlled laboratory environment to achieve fertilization.
4. **Embryo Culture:** The fertilized eggs (embryos) are cultured in the lab for a few days.
5. **Embryo Transfer:** The healthiest embryo(s) are transferred into the woman's uterus to establish pregnancy.

IVF is widely used to treat infertility caused by blocked fallopian tubes, low sperm count, ovulation disorders, or unexplained infertility. It has given hope to millions of couples worldwide who face challenges in conceiving naturally.

Quick Tip

IVF is one of the most common assisted reproductive technologies and has a higher success rate when performed at a younger age, as egg quality plays a major role in conception.

(b) Which bacteria converts milk into curd?

Solution:
The bacteria that convert milk into curd is **Lactobacillus**.
Step 1: Process of Conversion.
When a small amount of curd is added to warm milk, it introduces millions of Lactobacillus bacteria. These bacteria multiply rapidly in the warm environment.
Step 2: Chemical Reaction.
Lactobacillus produces **lactic acid** by fermentation of lactose (the sugar present in milk). This lactic acid lowers the pH of milk and causes the milk proteins (casein) to coagulate, resulting in the thick, sour substance known as curd.
Step 3: Importance.
The formation of curd improves digestibility, enhances the vitamin B content, and provides probiotics that are beneficial for gut health.

Quick Tip

Lactobacillus is the main bacteria used in curd formation. Remember: lactose (milk sugar) $\rightarrow$ lactic acid (by Lactobacillus) $\rightarrow$ curd.

(c) What is the difference between sugar of RNA and DNA?

Solution:
The main difference between RNA and DNA lies in the type of sugar molecule present in their nucleotides. Both nucleic acids contain a five-carbon sugar, but their structures vary slightly, which leads to differences in their properties and biological roles.
Step 1: Sugar in RNA.
In RNA (**Ribonucleic Acid**), the sugar is called **ribose**. Ribose is a pentose sugar with the following structural feature:

At the 2' carbon atom, ribose has a hydroxyl group (-OH).

This hydroxyl group makes RNA chemically more reactive and less stable under alkaline conditions. As a result, RNA molecules are usually single-stranded and perform temporary roles in cells such as protein synthesis (mRNA, tRNA, rRNA).

Step 2: Sugar in DNA.
In DNA (**Deoxyribonucleic Acid**), the sugar is called **deoxyribose**. The difference is:

At the 2' carbon atom, deoxyribose has only a hydrogen atom (-H) instead of a hydroxyl group.

This small change makes DNA chemically more stable, less reactive, and well-suited for long-term storage of genetic information.
Step 3: Functional Implications.

- The presence of the extra $-OH$ group in ribose makes RNA more prone to hydrolysis, limiting its stability. This is why RNA is usually short-lived and functions as an intermediary in the flow of genetic information.
- The absence of the $-OH$ group in deoxyribose provides DNA with higher resistance to enzymatic breakdown and chemical damage. This stability is critical for DNA's role in storing genetic information across generations.

Step 4: Summary of Difference.

RNA sugar: Ribose $\rightarrow$ has -OH at 2' carbon.

DNA sugar: Deoxyribose $\rightarrow$ has -H at 2' carbon.

Quick Tip

A simple memory trick: "R" in RNA stands for Ribose (with extra $-OH$), while "D" in DNA stands for Deoxy (without oxygen at 2'-carbon).

(d) Who is known as the father of 'Green Revolution' in India?

Solution:
The father of the **Green Revolution in India** is **Dr. M. S. Swaminathan**.
Step 1: Understanding the Green Revolution.
The Green Revolution refers to the period during the 1960s and 1970s when agriculture in India was transformed by the introduction of high-yielding varieties (HYVs) of seeds, chemical fertilizers, pesticides, irrigation methods, and mechanization. This significantly increased agricultural productivity, especially in wheat and rice.
Step 2: Role of Dr. M. S. Swaminathan.
- Dr. Swaminathan, an Indian geneticist and agricultural scientist, collaborated with Norman Borlaug (known as the father of the global Green Revolution) to bring HYV seeds of wheat to India. - He adapted these seeds to Indian conditions and guided large-scale implementation. - With his vision and leadership, India moved from a food-deficient country to becoming self-sufficient in food grain production.
Step 3: Impact of the Green Revolution.
- Helped India overcome famine-like conditions during the 1960s. - Improved food security and reduced dependence on food imports. - Marked the beginning of modern scientific farming in

India. - However, it also led to environmental concerns such as soil degradation, excessive use of fertilizers and pesticides, and reduced crop diversity.

Step 4: Recognition.

For his immense contribution, Dr. M. S. Swaminathan is honored with the title **"Father of Green Revolution in India"**. He also received global recognition and awards for his service in agricultural development and food security.

> Quick Tip
>
> The global Father of Green Revolution is Norman Borlaug, while in India, it is Dr. M. S. Swaminathan.

(e) How many chromosomes are found in humans?

Solution:

Humans have a total of **46 chromosomes**, arranged in **23 pairs**. These chromosomes are thread-like structures made of DNA and proteins, located in the nucleus of each cell. They carry hereditary information that controls growth, development, and functioning of the body.

Step 1: Breakdown of Chromosomes.

- **22 pairs of autosomes (44 chromosomes):** These are responsible for most of the body's traits such as height, skin color, eye color, metabolism, etc. - **1 pair of sex chromosomes (2 chromosomes):** These determine the sex of the individual. - Females: XX - Males: XY

Step 2: Role of Chromosomes.

Each chromosome contains thousands of genes, which are functional units of heredity. These genes encode proteins that control almost every biological function. Errors in chromosome number or structure can lead to genetic disorders.

Step 3: Examples.

- Down Syndrome: Caused by the presence of an extra copy of chromosome 21 (Trisomy 21). - Turner Syndrome: Occurs when one X chromosome is missing in females (XO). - Klinefelter Syndrome: Occurs when males have an extra X chromosome (XXY).

Step 4: Importance.

Knowledge of human chromosomes is essential in genetics, medical research, forensic science, and understanding hereditary diseases.

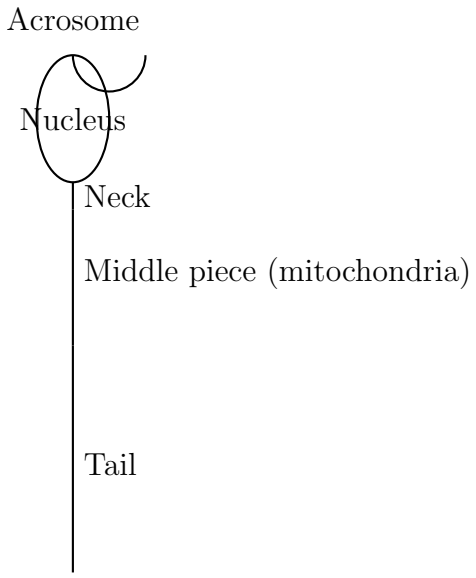
> Quick Tip
>
> Humans = 46 chromosomes (23 pairs). Out of these, 44 are autosomes and 2 are sex chromosomes. Mistakes in chromosome number cause genetic disorders.

3.(a) Draw only a labelled diagram of a sperm of man.

Solution:
The sperm is the male gamete in humans. It is a haploid cell (containing 23 chromosomes) produced in the testes by the process of spermatogenesis. A sperm has a unique structure that enables it to move actively and fertilize the female gamete (ovum).
Step 1: Structural divisions.
The human sperm is divided into three main parts:

1. **Head:** Contains the nucleus (haploid set of chromosomes) and is capped by the **acrosome**, which has enzymes (like hyaluronidase) that help in penetrating the egg.
2. **Middle piece:** Contains a large number of mitochondria which provide energy (ATP) for the movement of the sperm.
3. **Tail (Flagellum):** Long and whip-like, responsible for motility and helps the sperm swim towards the ovum.

Step 2: Diagram.
A labelled diagram of a human sperm is given below:



Step 3: Significance.
The human sperm is specialized for fertilization. The acrosome enzymes break down the outer layers of the ovum, while the mitochondria supply the energy for motility, ensuring the sperm can reach and fertilize the egg successfully.

> Quick Tip
>
> Remember: The sperm structure = Head (with nucleus + acrosome), Middle piece (with mitochondria), Tail (for movement).

(b) Describe Mendel's 'law of dominance'.

Solution:
Step 1: Background.
Gregor Mendel, the father of genetics, conducted experiments on garden pea plants (*Pisum sativum*) to study the inheritance of traits. From his experiments, he proposed three laws of inheritance:

1. Law of Dominance
2. Law of Segregation
3. Law of Independent Assortment

Here, we focus on the **Law of Dominance**.
Step 2: Statement of the law.
The Law of Dominance states that when two different alleles of a character are present in an organism (heterozygous condition), only one allele expresses itself (dominant), while the other allele remains masked (recessive).
Step 3: Example of Mendel's experiment.
Mendel crossed pure tall plants (TT) with pure dwarf plants (tt):

$$TT \times tt \quad \Rightarrow \quad F_1 \text{ generation: all } Tt$$

All plants in the F_1 generation were tall, showing that tallness (T) is dominant over dwarfness (t).
Step 4: Punnett Square.

	T	T
t	Tt	Tt
t	Tt	Tt

All F_1 hybrids are tall (Tt), proving that the dominant trait (Tallness) masks the recessive trait (Dwarfness).
Step 5: Importance of the law.

- Explains why certain traits appear in the F_1 generation and why recessive traits reappear in the F_2 generation.
- Foundation of classical genetics, used to predict inheritance patterns.
- Helps in plant and animal breeding to select desirable traits.

Step 6: Limitation.
The law of dominance does not explain incomplete dominance or codominance, where both alleles may express partially or equally (e.g., flower colour in snapdragon, ABO blood groups in humans).

Quick Tip

Law of Dominance: In heterozygotes, the dominant allele is expressed, and the recessive allele is masked. Example: Tall (T) is dominant over dwarf (t) in pea plants.

(c) Write the name of first amino acid and initiation codon for the initiation of protein synthesis.

Solution:
First Amino Acid: In **prokaryotes**, the first amino acid is **N-formylmethionine (fMet)**. In **eukaryotes**, the first amino acid is **Methionine (Met)**.
Initiation Codon: The initiation codon for protein synthesis in both prokaryotes and eukaryotes is:

AUG

This codon specifically codes for methionine, marking the start of translation. In some rare cases, codons GUG or UUG can also act as initiation codons in prokaryotes, but AUG is the universal and most common start codon.

Quick Tip

Remember: AUG is the universal start codon and methionine (or fMet in prokaryotes) is the first amino acid in protein synthesis.

(d) Write a short note on BT cotton.

Solution:
Bt cotton is a genetically modified (GM) variety of cotton that has been developed to resist insect pests, especially the **bollworm**, which causes severe damage to cotton crops.
Step 1: Development.
The gene for **Cry protein** was taken from the bacterium *Bacillus thuringiensis* (Bt). This gene was inserted into cotton plants through genetic engineering.
Step 2: Mode of Action.
The Cry protein, when ingested by insect larvae, gets activated in their alkaline gut. The protein binds to the gut lining, creating pores, which ultimately leads to the death of the pest.
Step 3: Benefits.

- Provides resistance against bollworm and other insect pests.
- Reduces dependence on chemical pesticides, making cultivation more eco-friendly.
- Increases cotton yield and reduces crop losses.

Step 4: Limitations.
Excessive and continuous use of Bt cotton may lead to the development of resistance in pests. It also raises concerns about biodiversity and ecological balance.

Quick Tip

Bt cotton contains a bacterial gene producing Cry protein that kills pests like bollworms, reducing pesticide use and improving yield.

(e) What are active and passive immunity?

Solution:

Step 1: Understanding Immunity.
Immunity is the ability of the body to resist infections or diseases by producing defense mechanisms. It is broadly classified into **active immunity** and **passive immunity**.
Step 2: Active Immunity.
Active immunity is the type of immunity in which an individual's own immune system is stimulated to produce antibodies and memory cells against a pathogen.

- It develops either after a natural infection (e.g., recovery from chickenpox) or through vaccination (e.g., polio vaccine).
- It provides **long-lasting protection**, often for years or even a lifetime.
- It takes time to develop since the body must first recognize the antigen and then produce the immune response.

Step 3: Passive Immunity.
Passive immunity is the type of immunity in which pre-formed antibodies are transferred into an individual's body, rather than being produced by the person's own immune system.

- Examples include transfer of maternal antibodies from mother to child through placenta or breast milk, and injection of antiserum or antivenom.
- It provides **immediate protection** but is usually **short-lived**, lasting only for weeks or months.
- No memory cells are formed, so there is no long-term immunity.

Comparison Table:

Feature	Active Immunity	Passive Immunity
Source	Produced by the body itself	Pre-formed antibodies from outside
Onset	Slow (takes time to develop)	Immediate protection
Duration	Long-lasting (years/lifetime)	Short-lived (weeks/months)
Memory Cells	Formed	Not formed
Examples	Vaccination, recovery from infection	Maternal antibodies, antivenom

Quick Tip

Active immunity = body "actively" makes its own antibodies (long-term). Passive immunity = body "passively" receives ready-made antibodies (short-term).

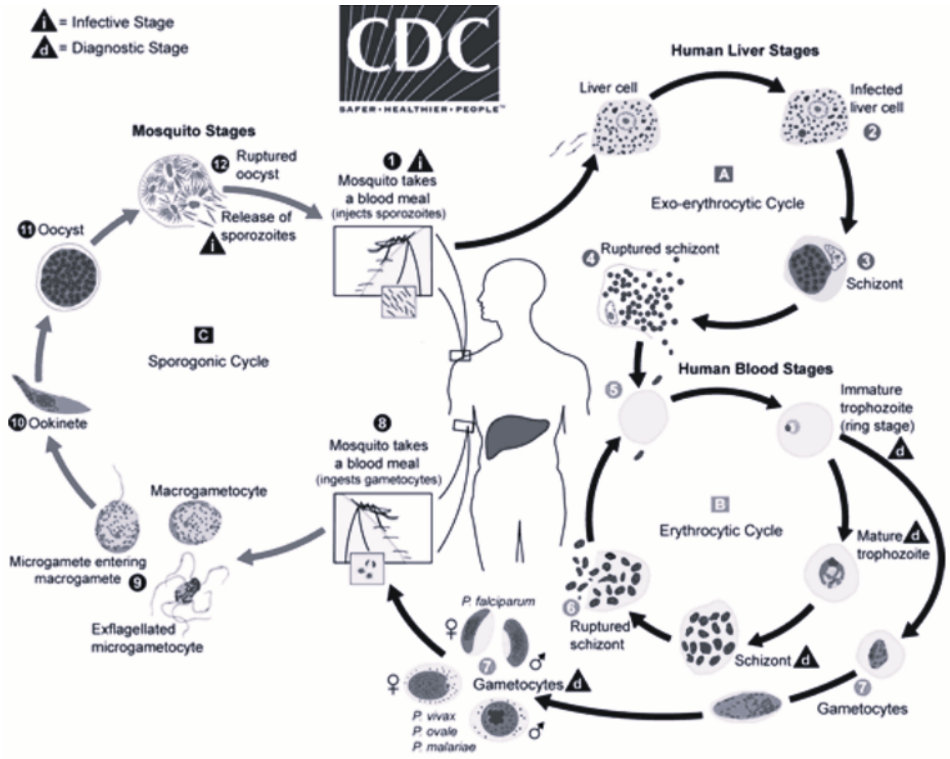
4.(a) Draw a clean and labelled diagram of the life-cycle of malaria parasite.

Solution:
The malaria parasite is caused by *Plasmodium* (e.g., *Plasmodium vivax*, *Plasmodium falciparum*). Its life cycle involves two hosts: **human (primary host)** and **female Anopheles mosquito (secondary host)**.
Step 1: Stages in human host.

1. Sporozoites are injected into the blood by the bite of female Anopheles mosquito.
2. They travel to the liver, undergo schizogony, and form merozoites.
3. Merozoites infect RBCs, multiply, and rupture them, releasing toxins (causing fever and chills).
4. Some merozoites develop into gametocytes.

Step 2: Stages in mosquito host.

1. When the mosquito bites an infected person, gametocytes enter its gut.
2. Male and female gametes fuse to form zygote.
3. Zygote develops into ookinete → oocyst → thousands of sporozoites.
4. Sporozoites migrate to salivary glands, ready to infect another human.



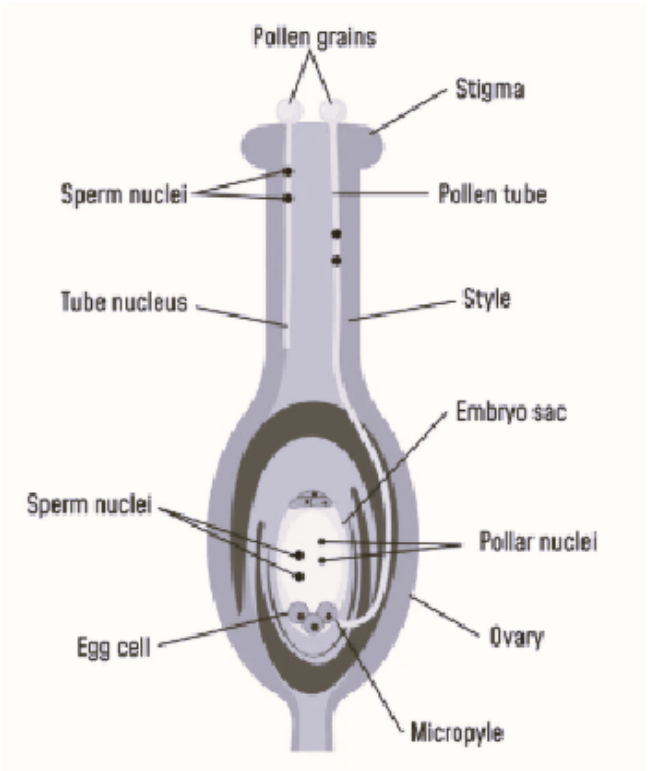
Quick Tip

Remember: Malaria parasite completes its life cycle in two hosts – **human (asexual cycle)** and **mosquito (sexual cycle)**.

(b) Draw a labelled diagram of an ovule including embryo-sac in a flowering plant.

Solution:
The ovule is the female reproductive structure in flowering plants. It develops into the seed after fertilization. Inside the ovule lies the embryo sac (female gametophyte), which plays a key role in double fertilization.
Step 1: Main parts of the ovule.

1. **Funicle:** Stalk that attaches ovule to placenta.
2. **Hilum:** Junction between ovule and funicle.
3. **Integuments:** Protective layers covering the nucellus.
4. **Micropyle:** Small opening through which pollen tube enters.
5. **Nucellus:** Nutrition tissue for the embryo sac.
6. **Embryo sac:** Female gametophyte with 7 cells (1 egg cell, 2 synergids, 3 antipodals, and 1 central cell with 2 polar nuclei).



Step 3: Significance.
The embryo sac is the site of double fertilization:

- One male gamete fertilizes the egg → zygote.
- Other male gamete fuses with two polar nuclei → triploid endosperm.

Quick Tip

Ovule + embryo sac = female gametophyte of flowering plants. It ensures double fertilization, a unique feature of angiosperms.

(c) Describe the role of any three micro-organisms in the production of household food materials.

Solution:
Micro-organisms play an important role in the production of various household food materials. They are used in fermentation processes which improve taste, texture, aroma, and nutritional value of food. Here are three important examples:
1. Lactobacillus (Bacteria):
- Found in curd (yogurt) preparation. - Converts lactose sugar present in milk into lactic acid. - The lactic acid gives curd its sour taste and thick texture. - Improves digestion and provides beneficial probiotics to the human gut.
2. Saccharomyces cerevisiae (Yeast):
- Commonly called **baker's yeast**. - Used in bread, cakes, and alcoholic beverage production. - During fermentation, yeast converts sugars into alcohol and carbon dioxide. - The carbon dioxide gas makes bread fluffy and soft, while in beer and wine production, alcohol is the main product.
3. Aspergillus oryzae (Fungus):
- Used in fermentation of soybeans to make soy sauce. - Produces enzymes that break down complex molecules into simple, flavorful compounds. - Enhances the taste and aroma of food.
Conclusion:
Thus, micro-organisms are vital in food industries as they help in making food more nutritious, easily digestible, and palatable.

Quick Tip

Remember: Lactobacillus → curd, Yeast → bread & alcohol, Aspergillus → soy sauce. Microbes are friends of humans in food processing.

(d) Write short notes on adaptation of desert plants and animals.

Solution:
Desert plants and animals show special adaptations to survive in extremely hot and dry conditions with very little water. These adaptations help them conserve water and tolerate high temperatures.

Adaptations in Desert Plants (Xerophytes):

1. **Reduced leaves or spines:** e.g., cactus has spines instead of leaves to minimize water loss by transpiration.
2. **Thick stems:** Many plants (like cactus, Euphorbia) have succulent stems that store water.
3. **Waxy coating:** Stems and leaves are covered with a thick cuticle to reduce evaporation.
4. **Deep roots:** Some plants have very deep root systems to absorb underground water, while others have widespread shallow roots to absorb surface rainwater quickly.

Adaptations in Desert Animals:

1. **Water conservation:** Camels can survive for long periods without water; they store fat in their humps which is converted into water and energy.
2. **Nocturnal habits:** Many desert animals (rats, foxes, lizards) are active at night to avoid daytime heat.
3. **Burrowing:** Animals like desert rats live in burrows to escape the hot sun.
4. **Efficient kidneys:** They excrete highly concentrated urine to minimize water loss.

Conclusion:
Both desert plants and animals have evolved structural, physiological, and behavioral adaptations to survive in harsh desert climates.

> Quick Tip
>
> Plants adapt mainly by water storage and reduced transpiration, while animals adapt by water conservation and behavioral changes like nocturnal activity.

5.(a) What is pollination? Write a note on the types of pollination of plants.

Solution:
Step 1: Definition.
Pollination is the process of transfer of pollen grains from the anther (male reproductive part) to the stigma (female reproductive part) of a flower. It is a prerequisite for fertilization in flowering plants.
Step 2: Types of Pollination.
Pollination is of two main types:

1. **Self-pollination (Autogamy):** Transfer of pollen grains from the anther to the stigma of the same flower or another flower on the same plant.
   - Advantages: Ensures seed production when pollinators are absent.
   - Disadvantages: Reduces genetic diversity.

2. **Cross-pollination (Allogamy):** Transfer of pollen grains from the anther of one flower to the stigma of another flower on a different plant of the same species.
    - Advantages: Promotes genetic variation, adaptability, and hybrid vigor.
    - Disadvantages: Depends on external agents (wind, insects, water).

Step 3: Agents of Pollination.
Cross-pollination may occur through:

- **Abiotic agents:** Wind (anemophily), Water (hydrophily).
- **Biotic agents:** Insects (entomophily), Birds (ornithophily), Bats (chiropterophily), etc.

> Quick Tip
>
> Self-pollination ensures purity of traits, while cross-pollination ensures genetic diversity.

(b) Comment upon sewage treatment.

Solution:
Step 1: Introduction.
Sewage treatment is the process of removing contaminants, organic matter, and harmful microorganisms from wastewater before releasing it into the environment. It protects human health and maintains ecological balance.
Step 2: Stages of sewage treatment.

1. **Primary treatment:** Involves physical removal of large particles through filtration, sedimentation, and grit chambers. Removes about 30–40% of organic matter.
2. **Secondary treatment (Biological treatment):** Utilizes microbes (bacteria, fungi, protozoa) to decompose organic matter.
    - Aerobic process: Uses aeration tanks and activated sludge.
    - Anaerobic process: Produces biogas from sludge digestion.
3. **Tertiary treatment:** Advanced stage involving chemical treatment, disinfection (chlorination, UV light), and nutrient removal (phosphates, nitrates). Ensures water is safe for release.

Step 3: Importance.

- Prevents spread of water-borne diseases.
- Reduces environmental pollution.
- Produces useful by-products like biogas and treated water for irrigation.

Quick Tip

Remember the 3 stages of sewage treatment: **Primary (physical)**, **Secondary (biological)**, and **Tertiary (chemical/advanced)**.

(c) Describe any three important characteristics of population.

Solution:
A **population** refers to a group of individuals of the same species living in a defined geographical area and capable of interbreeding. Its study includes several important characteristics. Three of the most important are:

1. **Population Density:** It refers to the number of individuals of a species per unit area or volume. Population density indicates how crowded or sparse a population is. For example, the population density of tigers in India is calculated as the number of tigers per 100 sq. km.
2. **Natality (Birth Rate):** Natality is the rate at which new individuals are born in a population during a given period of time. It is a key factor contributing to population growth. High birth rate increases population size while low birth rate slows growth.
3. **Mortality (Death Rate):** Mortality is the rate at which individuals die in a population during a given period of time. Mortality reduces population size and is influenced by environmental factors, diseases, and predation.

Other Characteristics (for reference): Age distribution, sex ratio, dispersal, and growth patterns are also important in population studies.

Quick Tip

Population studies focus on density, birth rate, and death rate as core parameters. These determine how populations grow, decline, or remain stable.

(d) What is the role of RNA in protein synthesis?

Solution:
RNA plays a crucial role in the process of **protein synthesis** (translation). Different types of RNA participate in various steps of protein formation from DNA-encoded genetic information.

1. **mRNA (Messenger RNA):** It carries the genetic code transcribed from DNA to the ribosomes. The sequence of codons on mRNA determines the sequence of amino acids in the protein.

2. **tRNA (Transfer RNA):** It transports specific amino acids to the ribosome during protein synthesis. Each tRNA has an anticodon that pairs with the codon on mRNA, ensuring the correct sequence of amino acids is added.
3. **rRNA (Ribosomal RNA):** It forms the structural and functional core of ribosomes. rRNA helps in binding mRNA and tRNA together and catalyzes the formation of peptide bonds between amino acids.

Summary: mRNA provides the code, tRNA brings the amino acids, and rRNA ensures the assembly—together making the central dogma of biology ($DNA \rightarrow RNA \rightarrow Protein$) possible.

> Quick Tip
>
> Remember: mRNA = message carrier, tRNA = amino acid transporter, rRNA = ribosome builder. All three work together for protein synthesis.

6.(a) Comment upon the following:
Plasmid Vector

Solution:
A **plasmid vector** is a small, circular, double-stranded DNA molecule found in bacteria, separate from the chromosomal DNA. It can replicate independently and is widely used in genetic engineering as a vehicle to transfer foreign DNA into host cells.
Key Features of Plasmid Vectors:

1. **Origin of Replication (Ori):** Ensures independent replication inside host cells.
2. **Selectable Marker:** Genes such as antibiotic resistance markers (e.g., ampicillin resistance) help in identifying transformed cells.
3. **Cloning Sites:** Multiple Cloning Sites (MCS) allow easy insertion of foreign DNA.
4. **Size:** Small size makes them easy to manipulate in the laboratory.

Example: pBR322, pUC19.

> Quick Tip
>
> Plasmids are the most commonly used vectors in recombinant DNA technology due to their easy replication and modification.

(b) Comment upon the following:
Apomixis

Solution:
Apomixis is a form of asexual reproduction in plants that mimics sexual reproduction but does

not involve fertilization. Seeds are formed without the fusion of gametes, leading to offspring that are genetically identical to the parent.
Types of Apomixis:

1. **Adventive Embryony:** Embryos develop directly from somatic tissues like nucellus or integuments.
2. **Diplospory:** Embryo sac develops without meiosis, and the egg cell divides to form embryo without fertilization.
3. **Apospory:** Embryo sac develops from somatic cells instead of megaspore.

Significance: - Helps in producing uniform crops. - Maintains hybrid vigour without the need for repeated hybridization. - Used in plant breeding programs.

> Quick Tip
>
> Apomixis ensures seed formation without fertilization and is useful in producing genetically identical plants.

(c) Comment upon the following:
Biodiversity and its Conservation

Solution:
Biodiversity refers to the variety of life forms on Earth, including plants, animals, fungi, and microorganisms, along with the ecosystems they form. It includes:

- **Genetic Diversity:** Variation of genes within a species.
- **Species Diversity:** Variety of species within a region.
- **Ecosystem Diversity:** Variety of ecosystems (forests, deserts, wetlands, etc.).

Need for Conservation: - Maintains ecological balance. - Provides food, medicine, timber, and raw materials. - Ensures ecosystem services like pollination, oxygen production, and climate regulation.
Methods of Conservation:

1. **In-situ Conservation:** Protecting species in their natural habitats, e.g., national parks, wildlife sanctuaries, biosphere reserves.
2. **Ex-situ Conservation:** Protecting species outside their natural habitat, e.g., botanical gardens, zoos, seed banks, gene banks.

> Quick Tip
>
> Biodiversity conservation is essential for ecological stability, sustainable development, and survival of human civilization.

(d) Comment upon the following:
Recombinant DNA Technology

Solution:
Recombinant DNA Technology (rDNA technology) involves joining DNA molecules from different species and inserting them into a host organism to produce new genetic combinations. It is a powerful tool of modern biotechnology.
Steps of Recombinant DNA Technology:

1. **Isolation of DNA:** The desired gene is identified and isolated.
2. **Cutting of DNA:** Restriction enzymes are used to cut DNA at specific sites.
3. **Insertion into Vector:** The gene is inserted into a vector (plasmid).
4. **Transfer to Host:** The recombinant vector is introduced into a host organism (like bacteria).
5. **Expression and Cloning:** The host expresses the foreign gene and produces the desired protein in large amounts.

Applications: - Production of insulin, growth hormones, interferons. - Development of genetically modified crops (Bt cotton, Golden rice). - Gene therapy for treating genetic disorders.

> Quick Tip
>
> Recombinant DNA technology revolutionized biotechnology by allowing scientists to alter genetic makeup for medical, agricultural, and industrial benefits.

7.(a) Comment upon any three chromosomal aberrations (syndrome) found in humans.

Solution:
Chromosomal aberrations are structural or numerical changes in chromosomes that cause genetic disorders in humans. These may involve extra, missing, or altered chromosomes. Three common syndromes are:
1. Down's Syndrome (Trisomy 21):

- Caused by the presence of an extra copy of chromosome 21 (2n=47).
- Symptoms: Mental retardation, short stature, broad forehead, flattened nose, short neck, slanting eyes, and heart defects.
- Frequency: Common in children born to mothers above 35 years.

2. Turner's Syndrome (Monosomy X):

- Occurs due to absence of one X chromosome in females (45, XO).
- Symptoms: Short stature, webbed neck, underdeveloped ovaries, sterility, and lack of secondary sexual characters.
- Affects only females.

3. Klinefelter's Syndrome:

- Occurs in males having an extra X chromosome (47, XXY).
- Symptoms: Tall stature, underdeveloped testes, sterility, enlarged breasts (gynecomastia), and reduced facial hair.
- Affects only males.

Conclusion:
These syndromes arise due to non-disjunction of chromosomes during gamete formation. They significantly affect physical and mental development.

> Quick Tip
>
> Remember: Down's = Trisomy 21, Turner's = 45, XO, Klinefelter's = 47, XXY. Easy to recall with numbers.

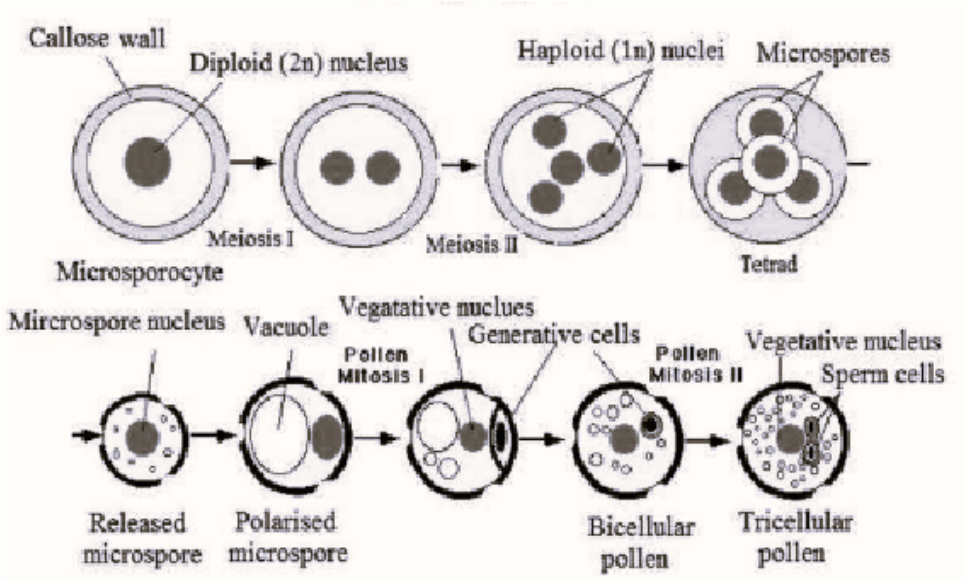
OR

(b) Describe the microsporogenesis in flowering plants with suitable diagram.

Solution:
Step 1: Definition.
Microsporogenesis is the process of formation of microspores (pollen grains) from microspore mother cells (MMC) in the anther of flowering plants.
Step 2: Process.

1. In the anther, diploid microspore mother cells (2n) undergo meiosis.
2. Each MMC forms a tetrad of four haploid microspores (n).
3. These microspores separate and develop into pollen grains, which act as male gametophytes.

Step 3: Diagram.



Step 4: Significance.

- Ensures genetic variation due to meiosis.
- Produces pollen grains for fertilization.
- Essential step in sexual reproduction of angiosperms.

Quick Tip

Microsporogenesis = MMC (2n) $\xrightarrow{\text{meiosis}}$ microspore tetrad (n) $\rightarrow$ pollen grains.

8.(a) Describe the semiconservative method of DNA replication with diagram.

Solution:
Step 1: Definition.
The semiconservative method of DNA replication was proposed by Watson and Crick and experimentally proved by Meselson and Stahl (1958). According to this model, during replication each DNA molecule produces two daughter DNA molecules, each having:

- One original (parental) strand.
- One newly synthesized strand.

Thus, half (semi) of the parental DNA is conserved in each daughter molecule.
Step 2: Process of replication.

1. **Initiation:** The double helix unwinds at the origin with the help of helicase enzyme. Hydrogen bonds between bases break, forming replication forks.
2. **Elongation:** Each parental strand acts as a template. DNA polymerase adds complementary nucleotides (A with T, G with C).

- Leading strand is synthesized continuously.
- Lagging strand is synthesized discontinuously as Okazaki fragments.

3. **Termination:** Okazaki fragments are joined by DNA ligase, resulting in two identical DNA molecules.

Step 4: Proof by Meselson and Stahl.
They used ^{15}N heavy isotope in *E. coli* DNA and showed that after one generation, the DNA molecules had one parental (old) strand and one new strand, confirming the semiconservative nature.

> Quick Tip
>
> Semiconservative replication = one parental strand + one new strand. Remember: Meselson–Stahl experiment is key proof.

OR

(b) Write an essay on population explosion and its control measures.

Solution:
Step 1: Introduction.
Population explosion refers to the rapid and excessive increase in human population over a short period of time. It creates immense pressure on natural resources, food supply, housing, healthcare, education, and employment.
Step 2: Causes of population explosion.

- Decline in death rate due to medical advances.
- Improved sanitation and nutrition.
- Lack of family planning awareness.
- Early marriage and preference for large families in some societies.

Step 3: Effects of population explosion.

- Overcrowding in cities, slums, and unemployment.
- Pressure on food, water, housing, and energy resources.
- Environmental degradation (deforestation, pollution, loss of biodiversity).
- Poverty, illiteracy, and spread of diseases.

Step 4: Control measures.

1. **Family planning:** Promotion of contraceptives, small family norm (two-child policy).
2. **Education:** Creating awareness, especially among women, about reproductive health.

3. **Government policies:** Incentives for small families, strict measures against child marriages.
4. **Healthcare:** Easy access to medical facilities and maternal care.
5. **Economic development:** Poverty eradication and women empowerment.

Step 5: Conclusion.
Population explosion can be controlled only through a combination of education, awareness, and strong implementation of government policies. Sustainable development depends on balancing population growth with available resources.

Quick Tip

Population explosion = more people, fewer resources. Family planning, education, and women empowerment are the key control measures.

9.(a) Give an account on Biosphere Reserves and National Parks.

Solution:

1. Biosphere Reserves:

- Biosphere reserves are large areas of protected land meant for conservation of biodiversity, research, and sustainable use of natural resources.
- They protect the flora, fauna, and the traditional lifestyles of the tribal communities.
- Each biosphere reserve typically consists of three zones:
  1. **Core zone:** Strictly protected for conservation of wildlife.
  2. **Buffer zone:** Limited human activities like research, tourism, and education are allowed.
  3. **Transition zone:** Human settlements, agriculture, and other activities are permitted in a sustainable manner.
- Examples in India: Nilgiri Biosphere Reserve, Nanda Devi Biosphere Reserve, Sundarbans Biosphere Reserve.

2. National Parks:

- National Parks are protected areas for wildlife conservation, where hunting, grazing, forestry, and private land use are prohibited.
- Aim: To conserve endangered species, maintain ecological processes, and promote ecotourism.
- National Parks are managed by the government and are strictly reserved for the preservation of biodiversity.

- Examples in India: Jim Corbett National Park (Uttarakhand), Kaziranga National Park (Assam), Gir National Park (Gujarat).

Quick Tip

Biosphere Reserves = large areas with core, buffer, and transition zones. National Parks = strictly protected areas for wildlife conservation.

OR

(b) Define menstrual cycle. Give an account on hormonal interplay during it.

Solution:

Step 1: Definition.
The menstrual cycle is a series of cyclic changes occurring in the ovaries and uterus of females, usually lasting about 28 days, starting from the first day of menstruation. It prepares the female body for fertilization and pregnancy.
Step 2: Phases of menstrual cycle with hormonal control.

1. **Menstrual phase (Day 1–5):**
   - Shedding of the uterine lining (endometrium).
   - Caused by a sudden fall in progesterone and estrogen due to degeneration of corpus luteum.
2. **Follicular/Proliferative phase (Day 6–14):**
   - Follicle Stimulating Hormone (FSH) from pituitary stimulates ovarian follicles to grow.
   - Developing follicles secrete estrogen.
   - Estrogen repairs and thickens the endometrium.
3. **Ovulatory phase (Around Day 14):**
   - A surge in Luteinizing Hormone (LH) triggers ovulation (release of ovum from Graafian follicle).
4. **Luteal/Secretory phase (Day 15–28):**
   - Corpus luteum forms from the ruptured follicle and secretes progesterone.
   - Progesterone maintains the endometrium for possible implantation of embryo.
   - If fertilization does not occur, corpus luteum degenerates, progesterone and estrogen levels drop, and menstruation starts again.

Step 3: Significance.
The menstrual cycle ensures that the uterus is prepared every month for the implantation of a fertilized egg, making it essential for reproduction.

Quick Tip

FSH = follicle growth, Estrogen = endometrium repair, LH surge = ovulation, Progesterone = endometrium maintenance.
